TIØ4145 Corporate Finance
Sustainable finance

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Sources

- Exam material: these slides
- For additional reading (not necessary for exam), see:
  - *Principles of Sustainable Finance* by Dirk Schoenmaker and Willem Schramade
  - *Corporate Finance for Long-Term Value* by Dirk Schoenmaker and Willem Schramade
  - Selected references on the slides

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Outline

- Sustainability's challenges to corporates
- What is sustainability and why does it matter?
- Externalities in managing sustainable development
- Sustainability and finance
  - Double materiality
- Climate risk analysis
- Internalization of externalities: policy measures
- Sustainability reporting
  - Main reporting requirements
- Investing for long-term value creation
  - Quantification of social and environmental value
  - Social discounting
  - Adjusted NPV
- Extra material: Green bonds

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Learning objectives

- Define sustainability and **key sustainability related challenges** for businesses today
- Explain the concept of **externalities** in managing sustainable development within corporations
- Describe the **role of financial industry** in achieving sustainable development sustainability can impact corporate financial decisions
- Understand how **policy measures** can **internalize environmental externalities**
- Define **climate and transition risks**
- Review various policy tools available for sustainability efforts
- Learn about the importance and methods of **sustainability reporting for corporations**
- Discuss key components of the **CSRD and EU Taxonomy** and their role in classifying sustainable activities
- Define **materiality** in the context of sustainability
- Learn methods to **quantify social and environmental impacts** of business activities
- Discuss the importance of this quantification in **corporate decision-making**
- Understand the concept of **social discounting** in evaluating sustainable investments.
- Discuss **how to adjust NPV** for sustainability factors in investment appraisals.

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What is sustainable finance?

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What is sustainable finance?

- … investment process accounting for and promoting environmental and social factors
- … is about environmental, social and governance considerations in investment decisions.
- ... is the practice of directing capital flows toward projects and companies with responsible environmental and social practices
- ... an approach to financial decision-making that integrates environmental, social, and governance (ESG) considerations to optimize long-term economic value and social benefit.
- …ethical decision-making in business and investment

- Sustainability is (Brundtland report, 1987): " … development that meets the needs of the present without compromising the ability of future generations to meet their own needs."

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What is ESG?

- ESG stands for **Environmental, Social, and Governance:** three key factors used in **the framework** to evaluate the sustainability and ethical impact of an investment in a business or organization.

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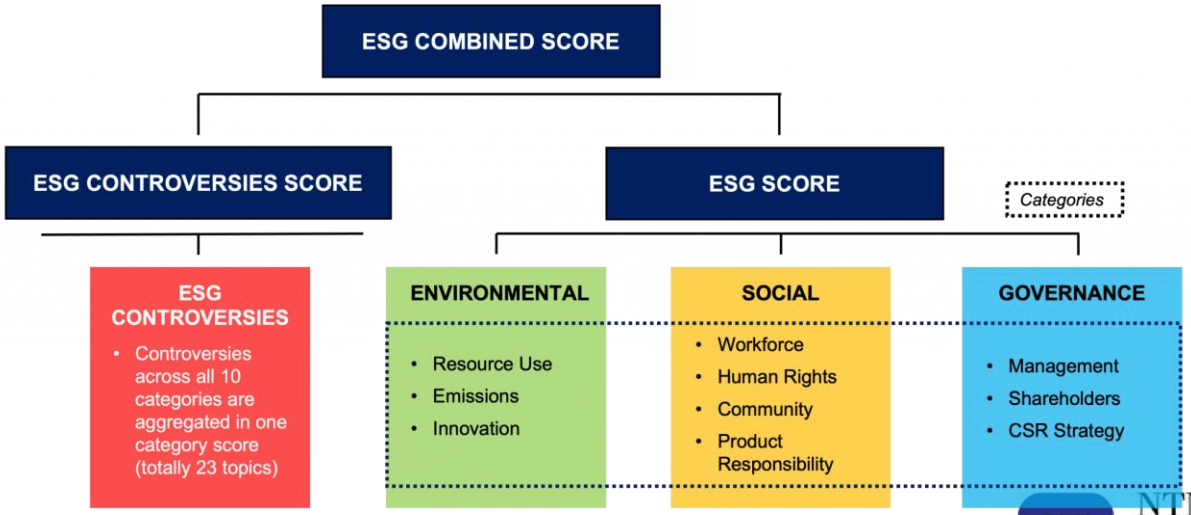
Available data on ESG mostly used

- Scores are percentile rank scores based on industry-relative performance
  - **ESG score**–Measures the company's ESG performance based on verifiable reported data in the public domain.
  - **ESG controversies score** –Measures the company's level of controversy based on negative media stories.
  - **ESGC score** –Overlays the ESG score with ESG controversies. Discounts the self-reported ESG score based on negative media stories.



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Challenges

- **ESG is a framework** that helps …
  - taking environmental, social and governance issues into account when trying to assess the potential risk-adjusted returns of an asset?
  - Investing in “ethical” or “green” or “sustainable” assets? But what is green?
- **Two completely different meanings!**
  - **ESG as an Input into Investment Decisions:** ESG as part of the process to assess potential financial returns. ESG issues are treated as factors that could affect the risk-adjusted returns of an investment.
  - **ESG as an Output (or Goal):** ESG as an ethical objective. Investments are made to promote sustainability and positive societal impact, even if the financial returns are less certain
- This conflict leads to myriad misunderstandings.
  - Is it OK to own a polluting Japanese manufacturer with terrible governance if these risks are considered less material than other drivers of returns?
  - Is it acceptable to invest in a promising clean energy startup with unproven financial performance, solely because its mission is to drive environmental and social progress?

Sustainability and traditional models

- Traditional economics and finance models often did not consider sustainability and environmental factors for several reasons:
  - **Historical Context**: Developed during periods when environmental resources seemed abundant and the focus was on industrial growth
  - **Lack of Intergenerational Accounting**: Insufficient emphasis on the welfare of future generations. Future costs and benefits were often discounted heavily, diminishing their importance in decision-making.
  - **Inadequate Predictive Models**: Lack of tools and data of predicting long-term environmental impact.
  - **Externalities:** Environmental costs are externalities not accounted for in private transactions.

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Externalities

- **Externality** - a consequence of an industrial or commercial activity which affects other parties without this being reflected in market prices.
  - Negative externalities - a polluter makes decisions based only on the direct cost of and profit opportunity from production and does not consider the indirect costs to those harmed by the pollution.
- Neoclassical economists recognized that the inefficiencies associated with externalities lead to “market failure.”
  - Private market–based decision making fails to maximize general welfare perspective. => Need for government intervention.
- Public good or free rider problem when there are no well-defined property rights
  - Public good - a commodity or service that is made available to all members of society (non-excludable and non-rivalrous)
- **Problem: costs and risk internalization**
  - The costs and risks from climate change are borne by the world at large, whereas there are few mechanisms to compel those who benefit from emission activity to internalize them.

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Different levels of sustainability actions

Sustainable finance: Key players

SUPRANATIONAL ORGANIZATIONS

Developing international agreements (e.g., Paris Agreement).

Coordinating multinational efforts, providing platforms for knowledge exchange (e.g., UN SDGs).

CORPORATES

High-emitting sectors: transition finance

Sustainable innovators (clean tech, renewables): recipients of green capital

Financial Sector: Channel capital and influence money flows

All companies: sustainability reporting, targets, sustainable supply chains

REGULATORY ACTORS

Implementing regulations to internalize externalities

Punish emitters (Carbon tax)

Support renewable energy investment

Improve sustainability reporting (CSRD)

Establishing protected areas to conserve biodiversity

CONSUMERS

Choosing eco-friendly products.

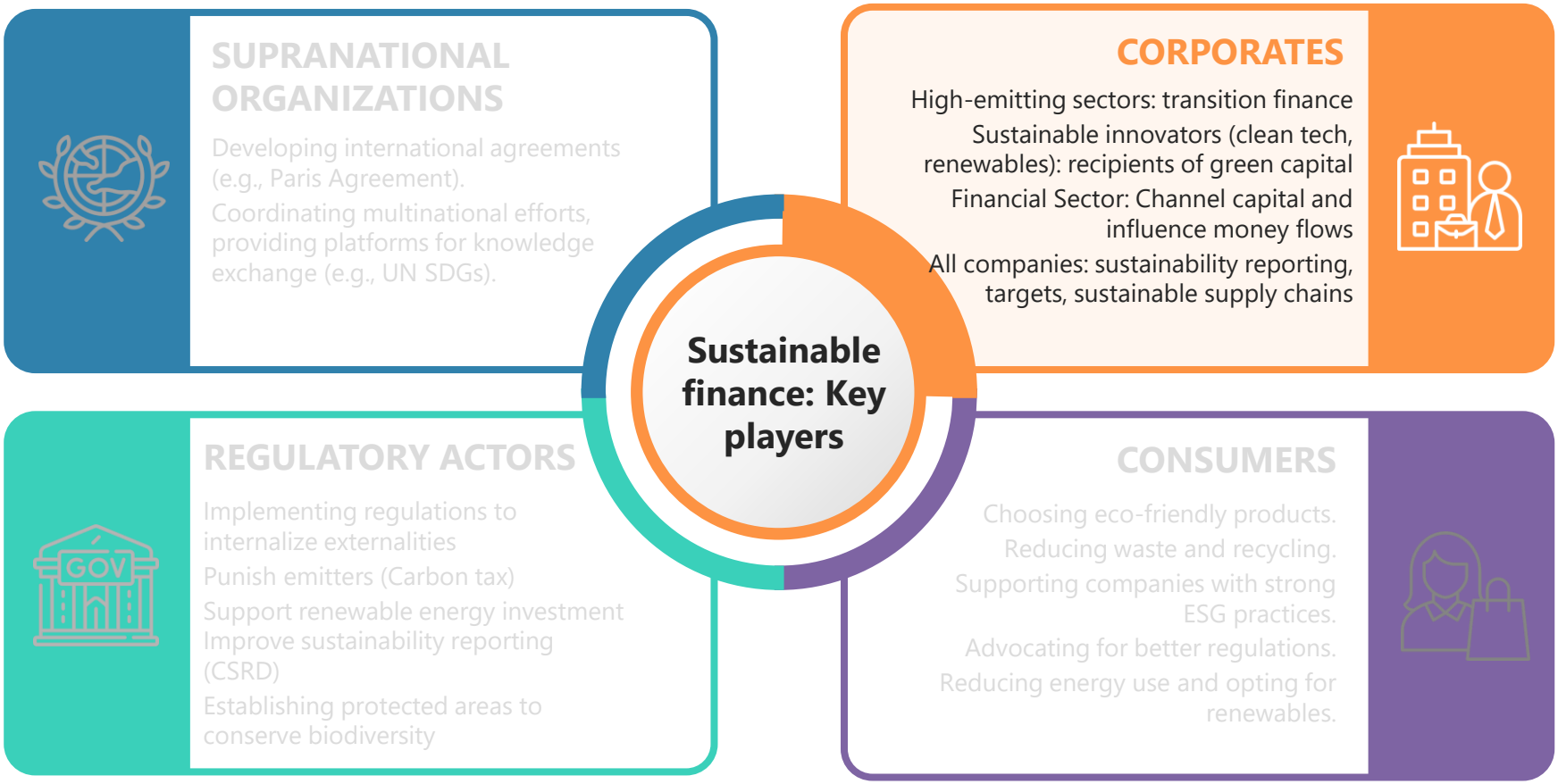
Reducing waste and recycling.

Supporting companies with strong ESG practices.

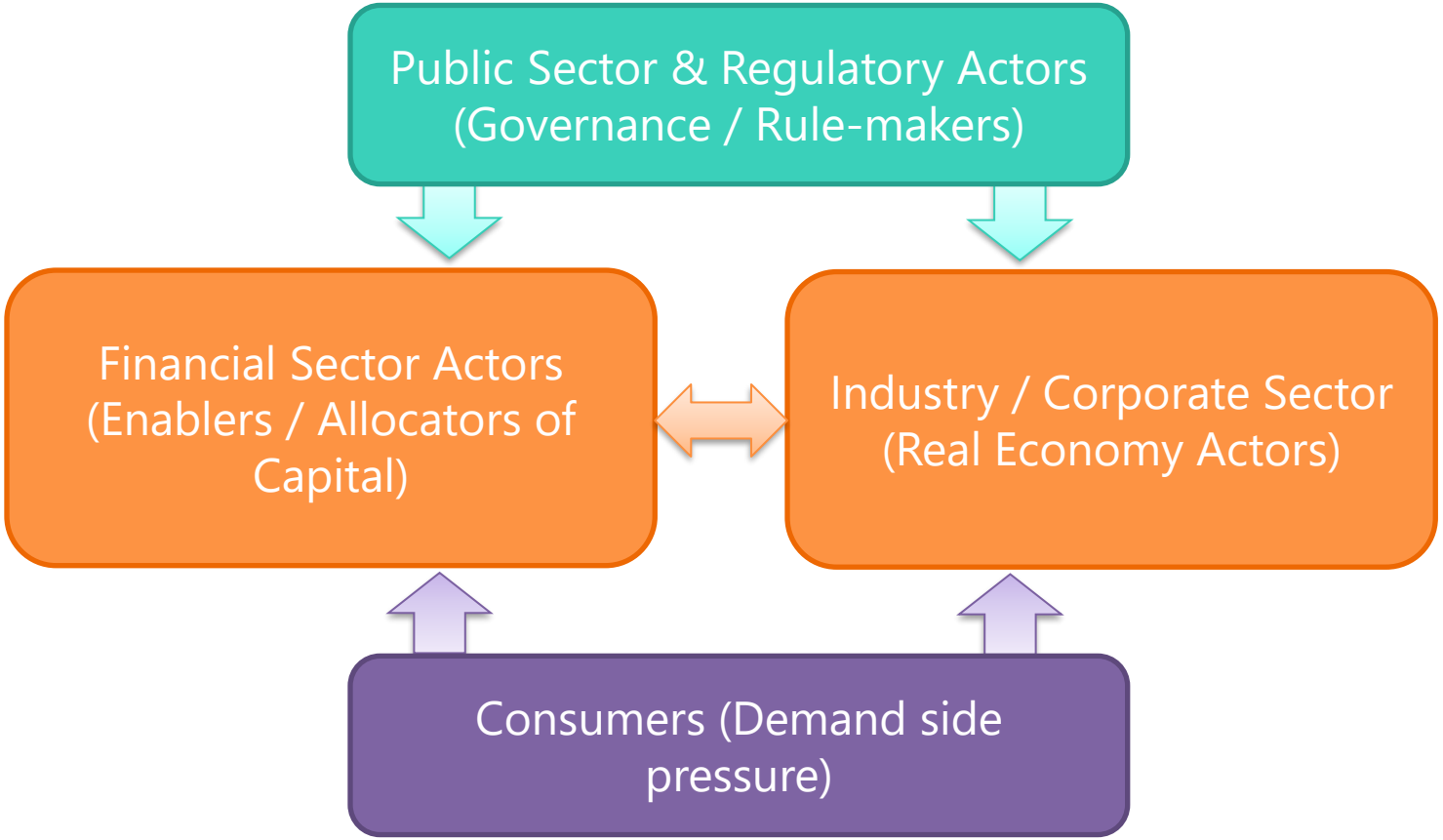
Advocating for better regulations.

Reducing energy use and opting for renewables.

Different levels of sustainability actions



This lecture: Sustainable finance actor map



How Industrials View Sustainability?

Direct Effects: Financial Impacts

- Impacts that affect the company's **cash flow or risk**:
  - **Costs & savings** (e.g., energy efficiency, green inputs, respond to ESG demand)
  - **Compliance & regulation** (e.g., carbon pricing, reporting obligations)
  - **Access to capital** (e.g., better terms on sustainability-linked loans)
- Often considered because they show up on the balance sheet.

Indirect Effects: Externalities

- Impacts that affect **society, nature, or future generations**. These may not immediately affect the company's financials
  - Emissions across the value chain, biodiversity loss, social inequality

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Role of financial industry?

Traditional Functions of Finance

- **Capital allocation**
  Produce information ex ante about possible investments, and allocate capital;
- **Corporate Governance**
  Monitor investments and exert corporate governance after providing finance;
- **Risk Management**
  Facilitate the trading, diversification and management of risk;
- **Market Facilitation**
  Ease the exchange of goods, services, and financial instruments.

Functions of Sustainable Finance

- **Mobilizing capital**
  Direct capital towards sustainable projects and companies.
- **Influence Corporate Behavior**
  Investors and lenders can exert pressure on companies to adopt sustainable practices.
- **Risk Management**
  Integrate ESG factors to manage risks associated with sustainable transitions.
- **Innovation in Financial Products**
  Developing financial instruments that incentivize sustainable business practices.

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How Financials View Sustainability?

Direct Effects: Financial Impacts

- Impacts that affect the **institution's financial performance or risk**:
  - **Credit risk** from lending to unsustainable sectors
  - **Market risk** from stranded assets or mispriced ESG factors
  - **Compliance & disclosure** obligations (e.g., EU Taxonomy, SFDR)
- Often prioritized because they influence profitability, capital requirements, or investor trust.

Indirect Effects: Externalities

- Impacts that affect **society, nature, or future generations**. These may not immediately affect the company's financials
  - Financed emissions, biodiversity loss from project finance, social harms from investment

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Framework for sustainable development

F — Financial and economic systems

S — Social impact

E — Environmental impact

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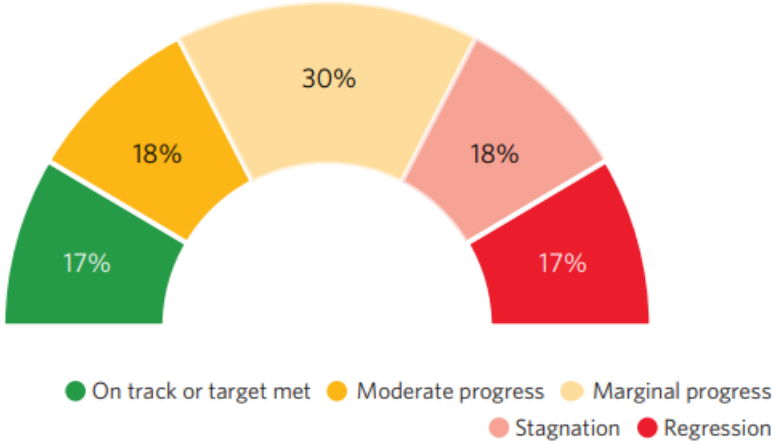
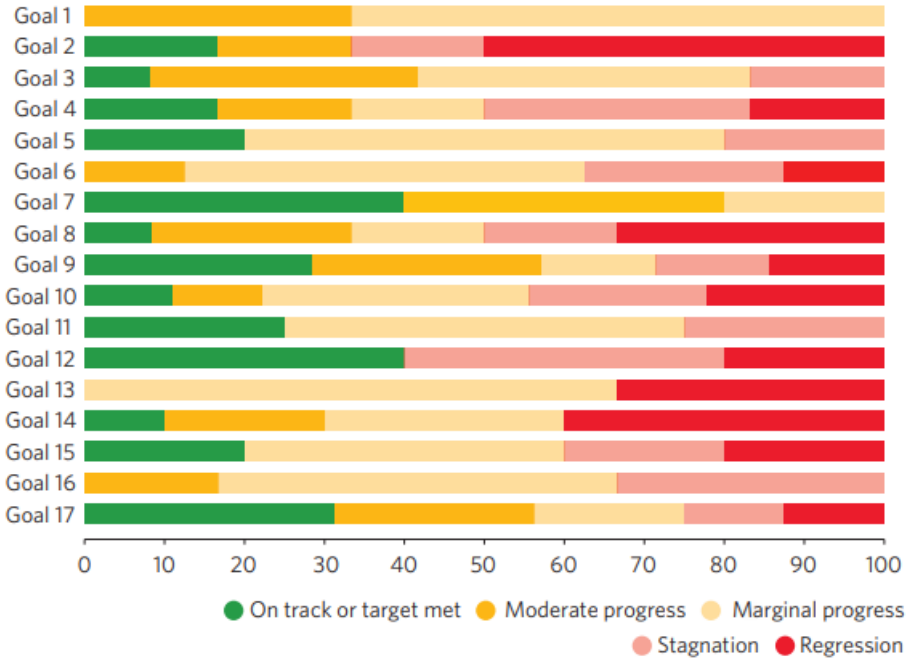
A reality check of the progress

Overall progress across targets based on 2015–2024 global aggregate data



Progress assessment for the 17 Goals based on assessed targets, by Goal (percentage)



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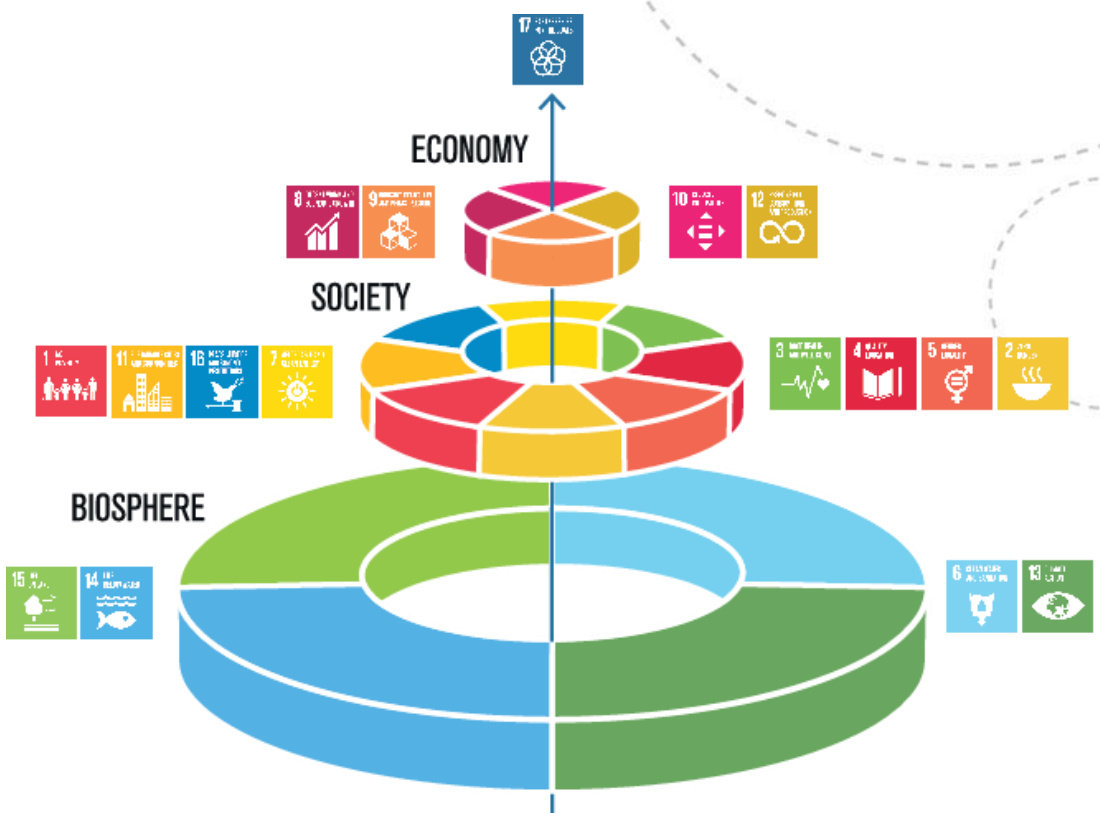
Framework for managing sustainable development

Financial return and risk: F

Impact on society: S

Impact on environment: E



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Three Stages of Sustainable Finance

Sustainable finance typology	Value created	Factors	Optimization	Horizon
Finance-as-usual	Shareholder value	F	Max F	Short term

Source: Bruegel. Note: F = financial value; S = social impact; E =environmental impact; T = total value

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From shareholder to stakeholder value

- The evolution highlights the broadening from shareholder value to stakeholder value (or triple bottom line: people, planet, profit).
  - In business, a stakeholder is any individual, group, or party that has an interest in an organization and the outcomes of its actions.
- The final stage looks at the creation of common good value.
- **Finance-as-usual**: In traditional finance, shareholder value is maximized by looking for the optimal financial return and risk combination.
  - Short-term focus (quarterly financial reporting and monthly/quarterly benchmarking of investment performance).
  - Role of business is to use its resources and engage in activities designed to increase its profits as long as it stays within the rules of the game.
  - Sustainability is the role of the government. Neoclassical economics tradition have argued that sustainability investments unnecessarily raise a firm's costs Creating a competitive disadvantage vis-à-vis competitors (Friedman, 1970)

- **Sustainable finance 1.0:** Profit maximization, while avoiding 'sin' stocks
  - In the social domain, they include, for example, companies that sell tobacco
  - In the environmental field, classic examples are waste dumping and whale hunting.

- **Sustainable finance 2.0:** : Internalization of externalities to avoid risk
  - Explicit incorporation of the negative social and environmental externalities into their decision-making.
  - Some of these externalities may be priced (e.g. a carbon tax) and/or impact negatively on an institution's reputation. Incorporating the externalities thus reduces the risk that financial investments become unviable.

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Sustainability and financial value

- Are sustainability initiatives value destroying?
- It depends on the definition of value
- However, even in monetary terms it is not so obvious
- Enhanced sustainability performance may lead to better returns and lower risks
  - Obtaining better resources
  - Higher quality employees
  - Better marketing of products and services
  - Risk mitigation, e.g. hedge against climate and transition risks
  - Improved capital flow due to reduction in information asymmetry (e.g. better reporting)

- Many firms now identify sustainability issues as **strategically important**
- Investors increasingly commit to incorporating ESG data in their **capital allocation process**

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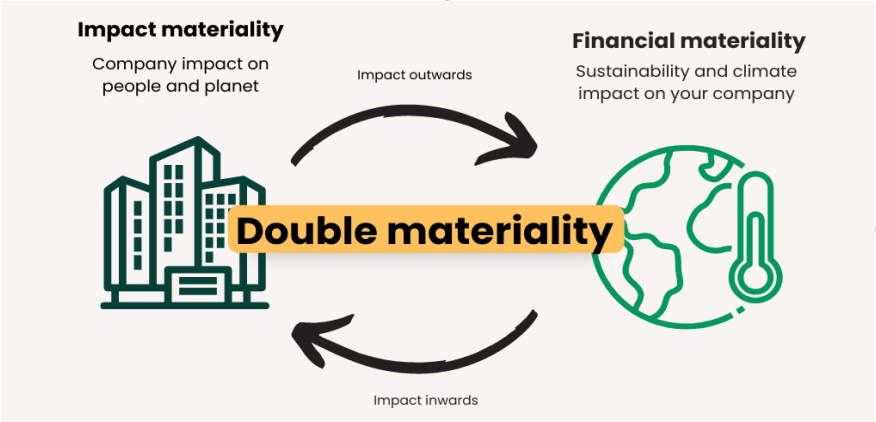
Double materiality

- **Materiality:** Traditional financial materiality refers to issues that impact a company's financial performance and risk profile (e.g., regulatory changes, market fluctuations)

- **Double Materiality:**
  - **Financial Materiality:** How ESG factors impact the company's financial health.
  - **Impact:** How the company's activities impact the environment and society.



Impact materiality

Impact

Sustainability matters, which "pertain to the undertaking's material actual or potential positive or negative **impacts on people or the environment** over the short-, medium- or long-term." (including products, services and business relationships). This indicates the company's contribution to sustainable development.

Considering scale, scope, irremediability and likelihood of impacts.

Financial materiality

Risk

Uncertain (environmental, social or governance) events or conditions, which "have a material negative influence or could reasonably be expected to have a material **negative influence on the undertaking's development**, financial position, financial performance, cash flows, access to finance or cost of capital over the short-, medium- or long-term".

Opportunity

Uncertain (environmental, social or governance) events or conditions, which "have a material positive influence or could reasonably be expected to have a material **positive influence on the undertaking's development**, financial position, financial performance, cash flows, access to finance or cost of capital over the short-, medium- or long-term".

Considering magnitude of financial effect and likelihood.

Source: ESRS 1 General Requirements

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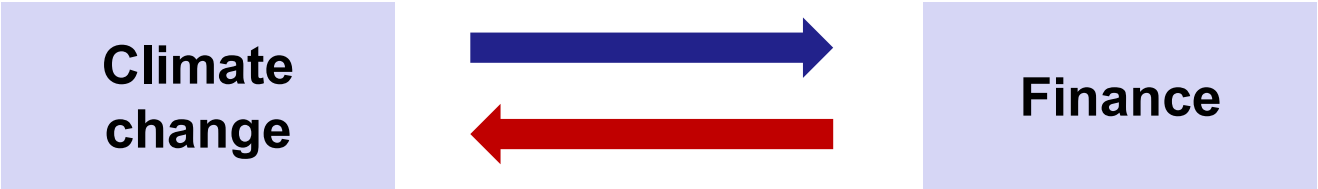
Example: Climate change

- Indirect impact of climate change on finance
  - Adjustment related to the mitigation of climate change, scenarios, transmission channels, estimates
- Direct impact of climate change on finance
  - Hazards on physical assets , scenarios, transmission channels, estimates



- Impact of finance on climate change
  - Risk perception, investment, coordination issues

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Physical risk versus transition risk

- **Climate physical risk:** refers to risk of damages to physical assets, natural capital and/or human lives resulting into economic output losses, as a result of climate related acute or chronic impact of climate change.
  - Based on available scientific information, in all scenarios such impacts are expected to increase. The magnitude grows with the level of warming across scenarios.
- **Climate transition risks:** changes in financial asset values that market players are not able to anticipate or hedge against:
  - Technological developments
  - Unexpected policy. Alternatively, even if the policy introduction is known, its impact and details come as unexpected
- **Stranded Assets**-"assets that lose their value".
  - This term was introduced for fossil fuel assets, which may become stranded due to government regulation (e.g., Carbon pricing) or technical change (e.g., reduced cost of solar photovoltaic (PV) or wind)

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Examples of materiality of climate risk

THE WALL STREET JOURNAL.

English Edition ▾ Print Edition | Video | Audio | Latest Headlines | More ▾

BUSINESS

PG&E: The First Climate-Change Bankruptcy, Probably Not the Last

The fast fall of PG&E after California's wildfires is a jolt for companies considering the uncertain risks of a warming planet

By Russell Gold Follow

Jan. 18, 2019 9:00 am ET

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Example from Norway

AUGUST 21, 2025 The GIST

How a 'perfect storm' triggered a marine heat wave and sent salmon lice into overdrive

by Bjerknes Centre for Climate Research

edited by Gaby Clark, reviewed by Robert Egan Editors' notes

Sun around the clock, clear skies, mirror-like water, and abnormal conditions in the North Atlantic. Thes…

On August 5, 2024, a marine heat wave began along the coast of Lofoten in Northern Norway. It lasted for 21 days, with sea temperatures measured at a record high. This caused salmon lice to thrive.

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Examples of materiality of transition risk

FINANCIAL TIMES

Dutch court orders Shell to accelerate emissions cuts

Ruling sets a precedent for other companies that could face similar lawsuits

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Examples in Norway

New salmon farming regulations postponed until at least 2026

6 June 2025

by Editorial Staff

Norway delays aquaculture reform as political compromise draws criticism from environmentalists.

Norway's centre-left government has postponed the introduction of new aquaculture regulations following a cross-party compromise that maintains the existing licensing system while initiating further studies. The move has drawn criticism from environmental groups and opposition parties, who argue that the measures fall short of protecting the country's struggling wild salmon population.

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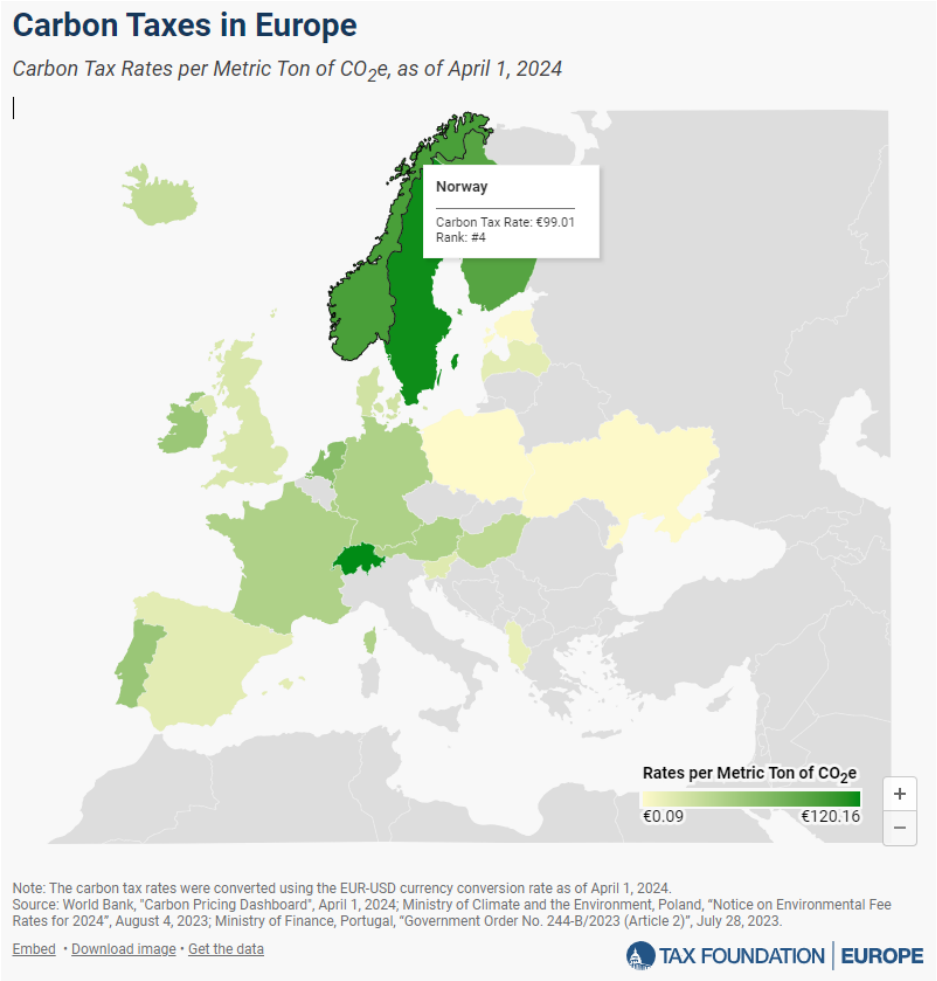
Examples of government interventions: Carbon tax



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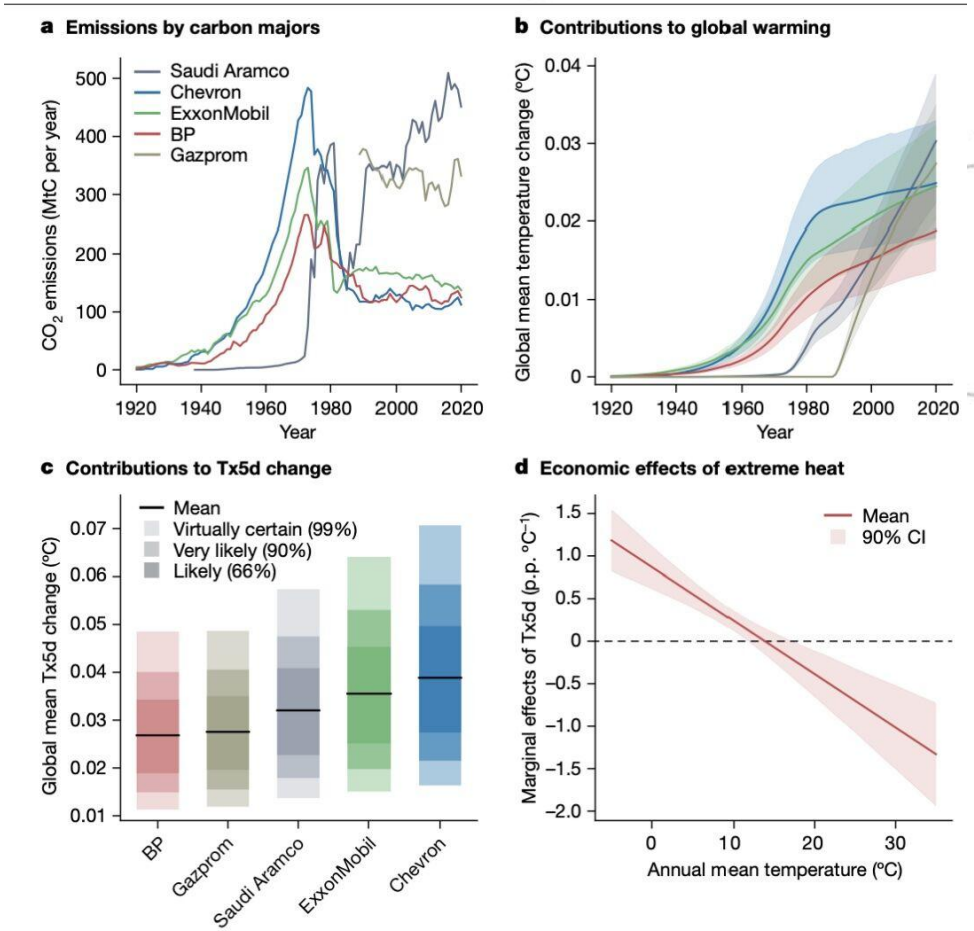
A case for targeted intervensions?

- **New study (Carbon majors and the scientific case for climate liability, Callahan & Mankin, 2025)** shows fossil fuel companies can now be *directly linked* to climate damages.
- **End-to-end framework**:
  company emissions → extreme heat → economic loss
- Results: **$28 trillion** in global damages traced to 111 producers.
  - Chevron alone: **$791B–$3.6T**Crucially: satisfies the strict **legal standard** (used in tobacco/asbestos cases).
- Implications: fossil fuel firms face **lawsuits, settlements, liabilities, insurance claims** at unprecedented scale.
- For investors: liability is now **scientifically quantifiable,** risks are financial as well as environmental.



Why are climate risks are difficult to analyze?

- **Forward-Looking Nature:**
  Climate risk develops over several decades, while financial markets often focus on short-term horizons. This mismatch creates challenges in aligning long-term climate effects with immediate investment decisions.
- **Non-Linearity:**
  Climate events, such as extreme weather conditions, do not follow predictable patterns or normal distributions, making it difficult to use historical data to predict future risks.
- **Ambiguity:**
  The scope and timing of climate change consequences are unclear, exacerbated by tipping points and domino effects that can lead to irreversible changes in the Earth's systems. The absence of clear probabilities for these events poses significant challenges for decision-making and financial risk assessments.
- **Complexity:**
  Addressing climate risk requires global coordination and depends on the behavior of various economic actors. These interactions often result in outcomes that cannot be predicted by analyzing individual behavior patterns alone.
- **Circularity:**
  Climate risk is closely intertwined with political decisions. Uncertainty about future political measures affects investor behavior, which in turn influences the success of these measures, creating a feedback loop.

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How do companies analyse climate risks?

Identify climate scenarios

Sector breakdown

Application to individual **companies based on sector exposure**

Effect on equity and debt

Macro-level scenarios

Insight into sectors and technologies

Company level assesment

Financial value

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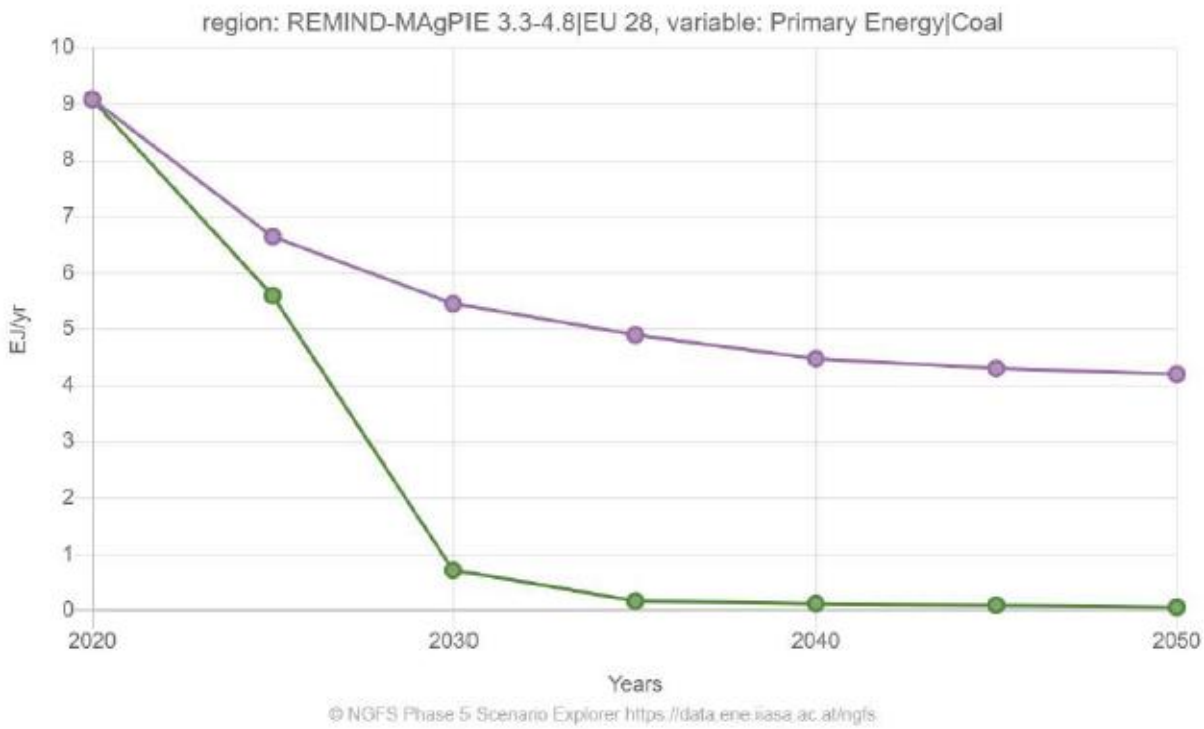
Example

- NGFS Scenario Explorer

REMIND-MAgPIE (Regional Model of Investments and Development - Model of Agricultural Production and Its Impact on the Environment) er en integrert vurderingsmodell som kombinerer en modell for energisystemer, økonomi og klima og en modell for arealbruk og jordbruksproduksjon.



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Risk Analysis?

- **Scenario Analysis ≠ Risk Analysis**
- **Climate scenarios** (e.g., NGFS, IPCC) are *narratives* about possible futures, they **do not assign probabilities**.
- Scenario analysis shows **exposure** of a portfolio or company under a given set of assumptions.
- **Risk analysis** requires attaching probabilities and distributions to outcomes — which scenarios *alone* cannot provide.

- Be careful: scenarios are **stress-tests, not forecasts**. They highlight vulnerabilities, but they don't tell you how likely those futures are.

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So far …

- …looking into what ESG factors affect risk and return and translating it in financial value
- View that sustainable investment is a “good deal”

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Challenges

The problem with sustainable investing:

Unhelpful definitions of “sustainable”

Unreliable ratings

Lack of comparability

Challenges in assessing the success of socially responsible investing

Difficulty of scaling up truly effective impact investing.

The problems with reporting:

Lack of mandates and auditing

Specious targets

Opaque supply chains

Complexity

Confusing information

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Where we are at the moment

- Sustainability understanding in the industry at the moment
  - **Extended scope** (Climate, extreme events risks, circular economy, social dimension)
  - **Important strategic choices**: Sustainability a problem or a solution? Competence, business, model, investment
  - **Compliance! Compliance! Compliance!**
    - EU Taxonomy,
    - CSRD (The Corporate Sustainability Reporting Directive),
    - TCFD (Task Force on Climate-Related Financial Disclosures),
    - TNFD (Task Force on Nature-Related Financial Disclosures).

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Regulatory landscape

- **EU's Green Deal** was launched in 2019 as a follow-up to the **Paris Agreement**. The main goal of the "Green Deal" is to make Europe climate-neutral by 2050.

- **EU Action Plan on Sustainable Finance** is part of the "Green Deal." The action plan is intended to promote investments in sustainable projects, among other things through strengthened sustainability reporting.

- This leads to the **Corporate Sustainability Reporting Directive (CSRD)** and the **accompanying European Sustainability Reporting Standards (ESRS)**

….and double materiality analysis is a central part of the CSRD.

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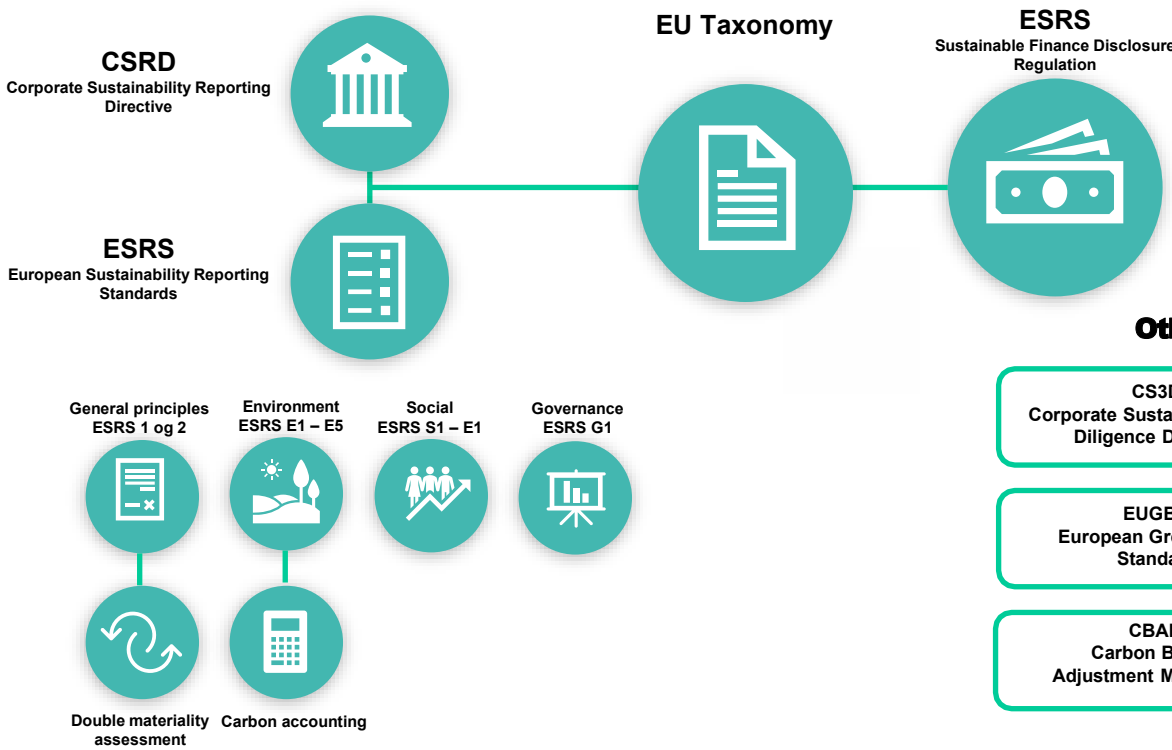
Regulation and legislation are mechanisms designed to ensure compliance with the action plan for sustainable finance



Other laws and regulations

- CS3D Corporate Sustainability Due Diligence Directive
- Changes in legislation related to investment advice, etc.
- EUGBS European Green Bond Standard
- EU Climate Benchmark Regulation
- CBAM Carbon Border Adjustment Mechanism
- EU Greenwashing Directive

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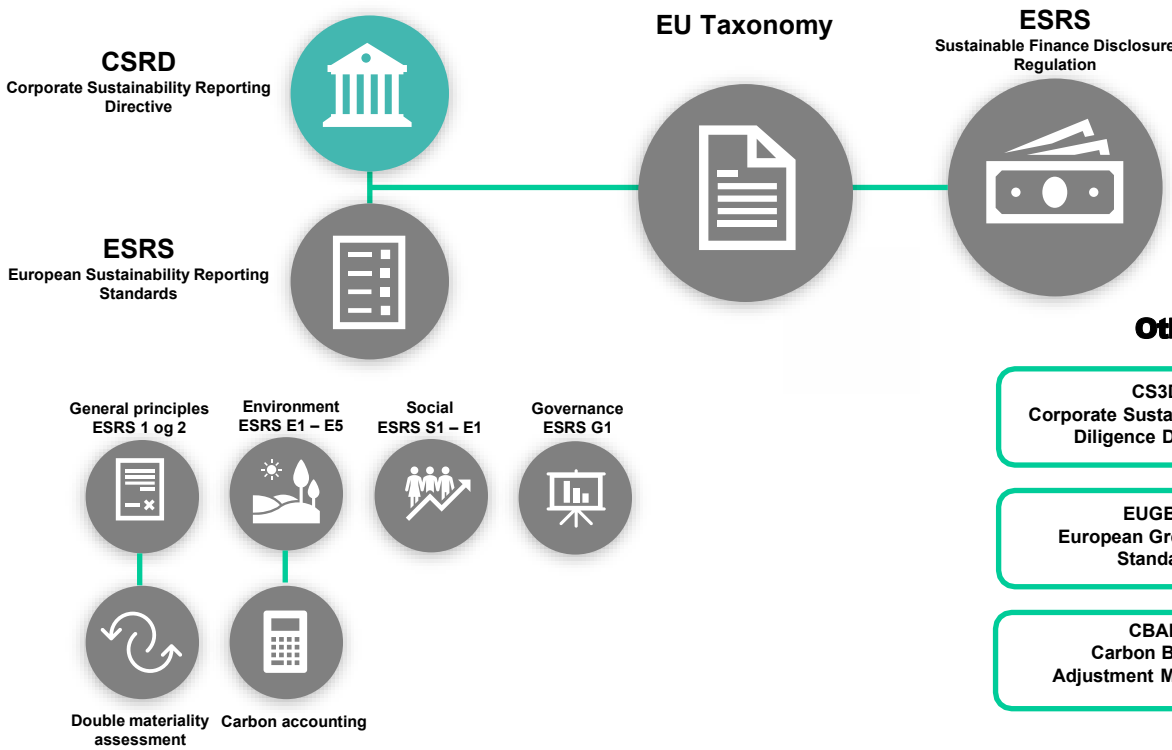

Other laws and regulations

- CS3D Corporate Sustainability Due Diligence Directive
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- EUGBS European Green Bond Standard
- EU Climate Benchmark Regulation
- CBAM Carbon Border Adjustment Mechanism
- EU Greenwashing Directive

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CSRD

- European Union's CSRD
  - On January 5, 2023, the European Union's Corporate Sustainability Reporting Directive (CSRD) took effect (EU-2022/2464). Under the CSRD, certain public companies will be required to annually complete and submit a report that complies with the European Sustainability Reporting Standards (ESRS). Under the current directive, the CSRD applies to companies with the following:
    - Over 250 employees
    - More than 40€ million in annual revenue
    - More than 20€ million in total assets
    - Publicly listed equities and have more than ten employees or 20€ million revenue.
    - International and non-EU companies with more than 150€ million annual revenue within the EU and which have at least one subsidiary or branch in the EU
- By the EU's estimates, 50,000 companies based in an EU member state or who do significant business within the EU will now be required to disclose ESG metrics annually. This is the most extensive enacted rule requiring ESG disclosures, affecting the largest volume of companies worldwide.

- Note! In the United States
  - No legal or regulatory schemes require any company, public or private, to make annual, public ESG disclosures.
  - In 2022, the Securities and exchange commission (SEC) released two proposed rule changes which would require foreign and domestic registrants to formally include certain climate-related information.
  - However, it is unclear if and when such disclosures will be mandated.

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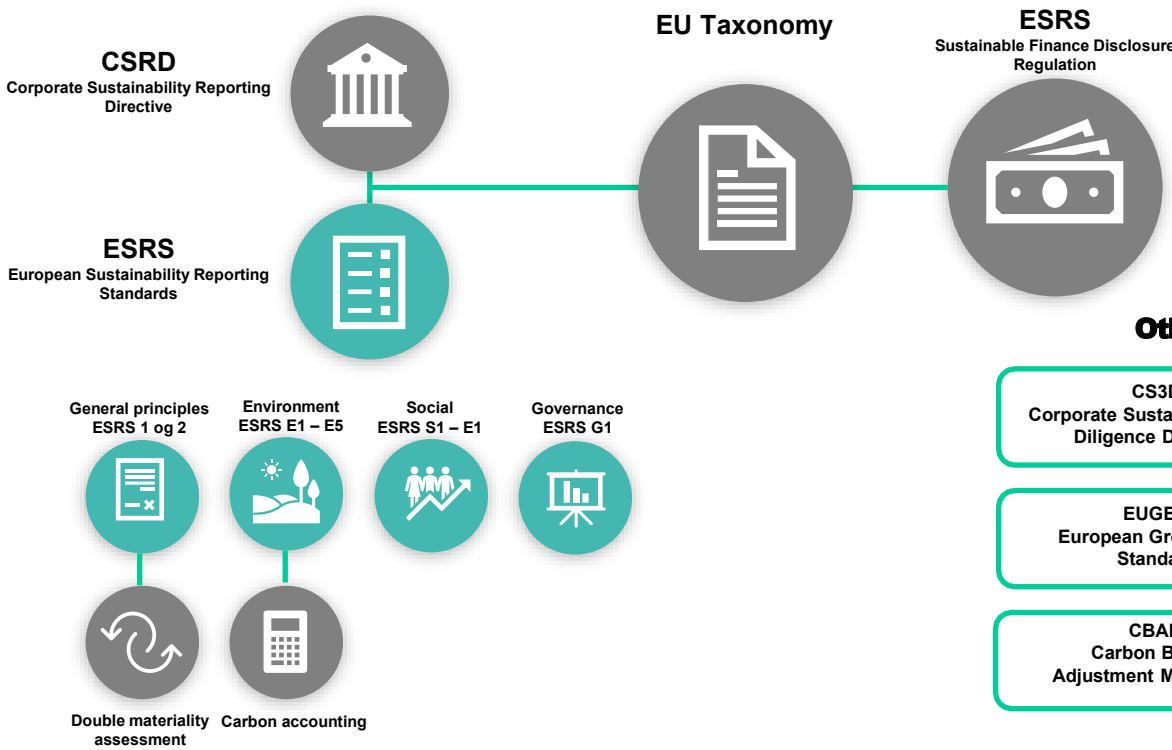
Regulation and legislation are mechanisms designed to ensure compliance with the action plan for sustainable finance



Other laws and regulations

- CS3D Corporate Sustainability Due Diligence Directive
- Changes in legislation related to investment advice, etc.
- EUGBS European Green Bond Standard
- EU Climate Benchmark Regulation
- CBAM Carbon Border Adjustment Mechanism
- EU Greenwashing Directive

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ESRS E1

CLIMATE CHANGE

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Strategy
- Disclosure Requirement E1-1 – Transition plan for climate change mitigation
- Disclosure Requirement related to ESRS 2 SBM-3 – Material impacts, risks and opportunities and their interaction with strategy and business model

Impact, risk and opportunity management
- Disclosure requirement related to ESRS 2 IRO-1 – Description of the processes to identify and assess material climate-related impacts, risks and opportunities
- Disclosure Requirement E1-2 – Policies related to climate change mitigation and adaptation
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Metrics and targets
- Disclosure Requirement E1-4 – Targets related to climate change mitigation and adaptation
- Disclosure Requirement E1-5 – Energy consumption and mix
- Disclosure Requirement E1-6 – Gross Scopes 1, 2, 3 and Total GHG emissions
- Disclosure Requirement E1-7 – GHG removals and GHG mitigation projects financed through carbon credits
- Disclosure Requirement E1-8 – Internal carbon pricing
- Disclosure Requirement E1-9 – Anticipated financial effects from material physical and transition risks and potential climate-related opportunities

Appendix A: Application Requirements
- ESRS 2 General disclosures

Strategy
- Disclosure Requirement E1-1 – Transition plan for climate change mitigation

Impact, risk and opportunity management
- Disclosure Requirement E1-2 – Policies related to climate change mitigation and adaptation
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ESRS S1

OWN WORKFORCE

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- Disclosure Requirement S1-3 – Processes to remediate negative impacts and channels for own workers to raise concerns
- Disclosure Requirement S1-4 – Taking action on material impacts on own workforce, and approaches to mitigating material risks and pursuing material opportunities related to own workforce, and effectiveness of those actions

Metrics and targets
- Disclosure Requirement S1-5 – Targets related to managing material negative impacts, advancing positive impacts, and managing material risks and opportunities
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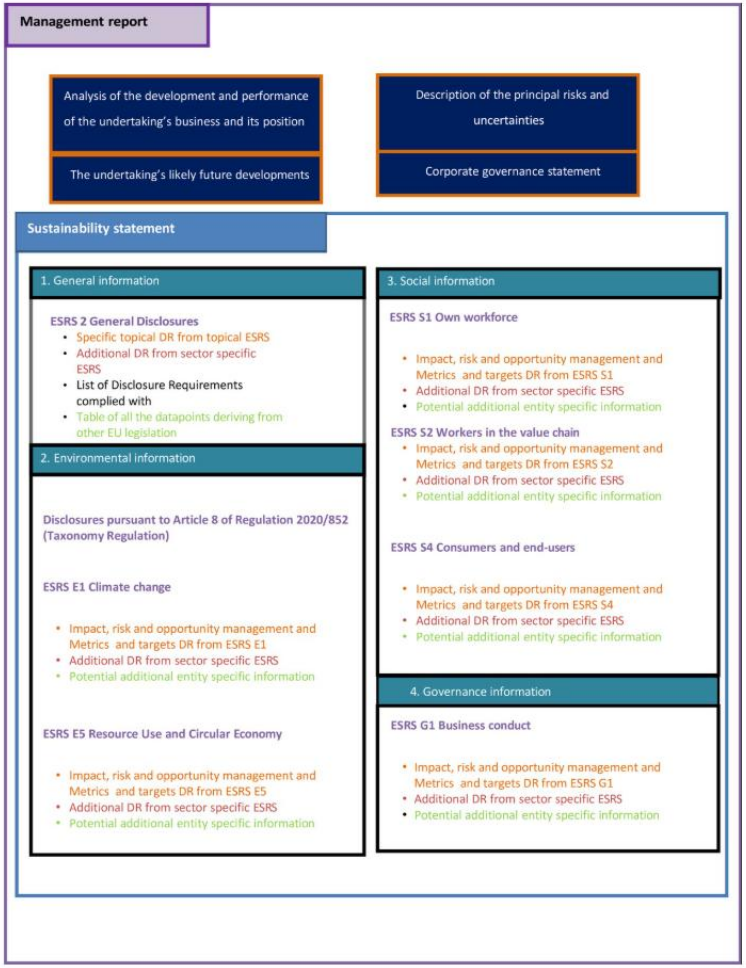
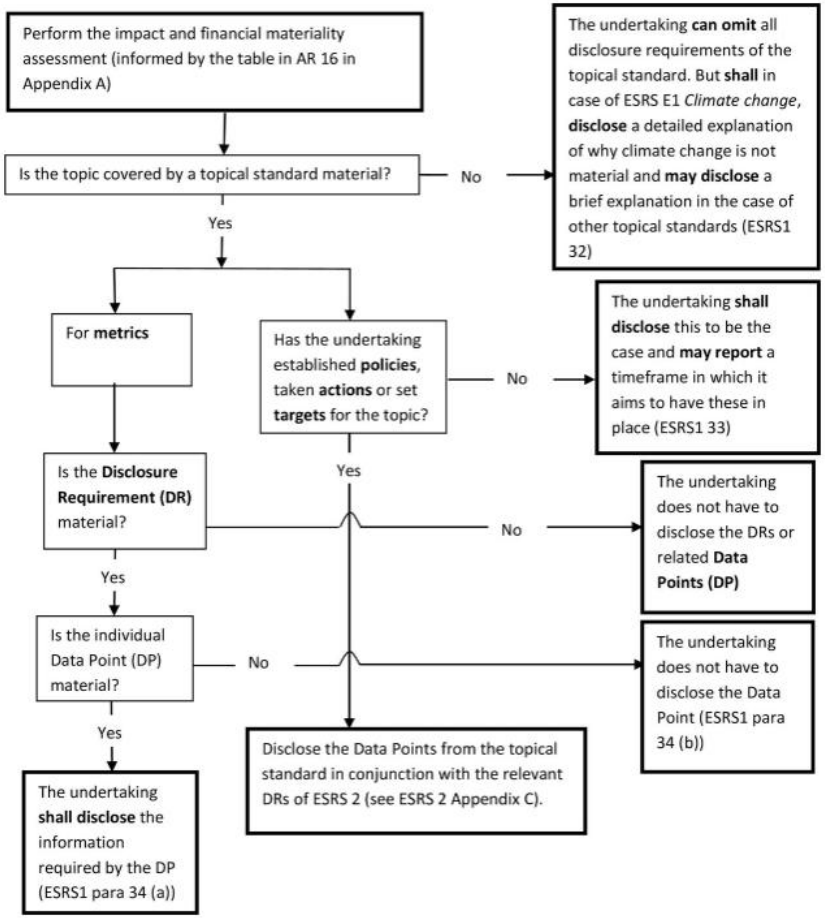
- Disclosure Requirement S1-7 – Characteristics of non-employee workers in the undertaking's own workforce
- Disclosure Requirement S1-8 – Collective bargaining coverage and social dialogue
- Disclosure Requirement S1-9 – Diversity metrics
- Disclosure Requirement S1-10 – Adequate wages
- Disclosure Requirement S1-11 – Social protection
- Disclosure Requirement S1-12– Persons with disabilities
- Disclosure Requirement S1-13 – Training and skills development metrics
- Disclosure Requirement S1-14 – Health and safety metrics
- Disclosure Requirement S1-15 – Work-life balance metrics
- Disclosure Requirement S1-16 – Compensation metrics (pay gap and total compensation)
- Disclosure Requirement S1-17 – Incidents, complaints and severe human rights impacts





https://eur-lex.europa.eu/legal-content/en/TXT/?uri=CELEX:32023R2772

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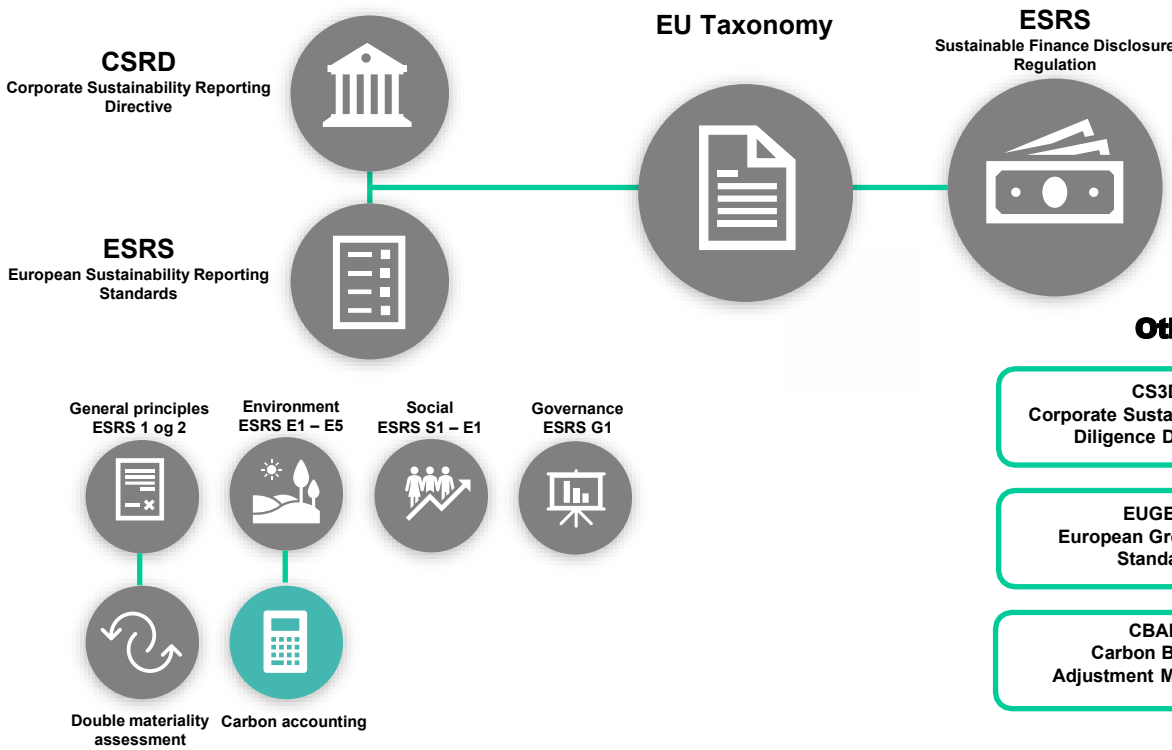

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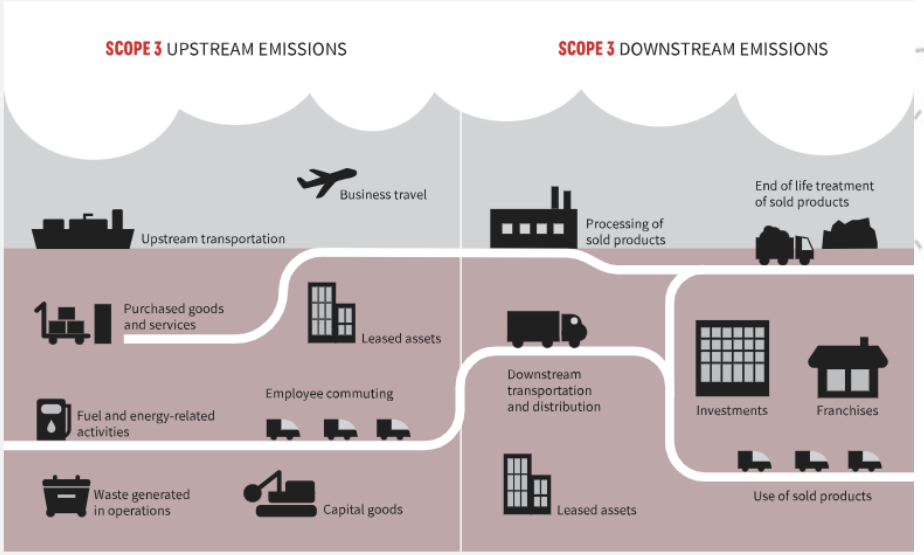
Carbon accounting

- **Which Emissions Must Be Reported?**
  - **Scope 1: Direct emissions** from sources owned or controlled by the reporting company (e.g., company facilities, company vehicles).
  - **Scope 2: Indirect emissions** from the generation of purchased or acquired electricity, steam, heating, or cooling consumed by the company.
  - **Scope 3: All other indirect emissions** in the value chain, upstream and downstream (e.g., purchased goods and services, transportation and distribution not controlled by the company, use of sold products, end-of-life treatment of products, etc.).
- **Important**: Under ESRS E1, companies must report on all three scopes **unless** they can justify that certain emission sources (especially some elements of Scope 3) are not material. If a company concludes a category is not material, it must briefly **explain the reasons** for that conclusion.

Exhibit 2: Scope 3 emissions are the most complex to measure



Source: Oliver Wyman analysis

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EU Taxonomy

- The EU Taxonomy is a classification system developed by the EU to identify and define economic activities that are considered environmentally sustainable.
- What is the scope of the Taxonomy Regulation requirements?
  - The Taxonomy Regulation will require the majority of large companies to report on the environmental sustainability of their economic activities (CSRD)

What is it?

- A list of economic activities and performance criteria for contribution to six environmental objectives

Environmental objectives

- Climate change mitigation
- Climate change adaptation
- Sustainable use and protection of water and marine resources
- Transition to a circular economy, waste prevention and recycling
- Pollution prevention and control
- Protection of healthy ecosystems

Substantial contribution of activities to at least one of the six environmental objectives as defined in the Regulation

1. Eligibility: activities must be included in the list of eligible ones

2. Threshold: metrics need to meet specific thresholds

3. DNHS: Do no significant harm to any of the other five objectives

Activities have to meet all three requirements

- As of 1 January 2022, has applies to the first two environmental objectives
  - From 1 January 2023, its scope of application also includes water and marine resources, the transition to a circular economy, pollution control, and biodiversity.

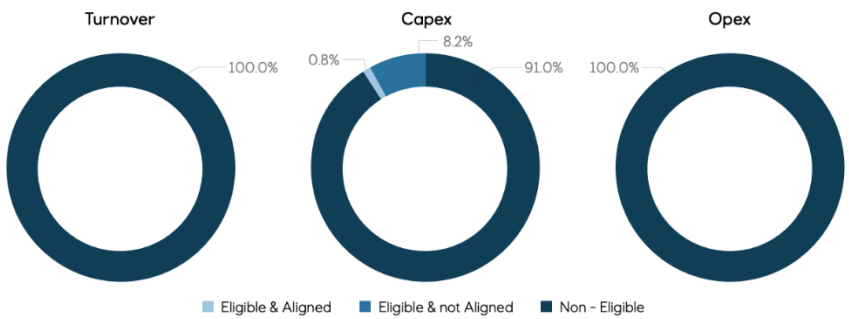
Example Taxonomy reporting - Equinor

Net Zero
In order to achieve its ambition to become a net-zero emissions company by 2050, Equinor undertakes emission reducing activities that are supporting the continued operation of oil and gas production. While these help Equinor towards its ambition, some of these activities (notably onshore electrification of offshore assets) are not eligible per the EU Taxonomy regulations and therefore are not visible in our eligibility scores.

The increase of 6.5% in the eligible capex mandatory KPI from 2022 primarily relates to investments in the solar Energy company BeGreen, Wind Lease in California and the Renewable company Rio Energy during the year.

Equinor's aligned activities in 2023 included in the mandatory KPIs above primarily consist of the offshore wind projects Hywind Tampen and Empire Wind, and an onshore solar project in Poland.

The investments in 2023 of BeGreen and the wind lease in California consist of acquisition costs related to future project developments which were not deemed material for alignment testing in 2023. The investment in the renewable company Rio Energy in the fourth quarter 2023 will be assessed for alignment in 2024.



DEFINITIONS

TAXONOMY **ELIGIBLE**: if the activity is described in the regulation*, irrespective of whether it complies with the technical screening criteria

TAXONOMY **ALIGNED**: if the activity contributes substantially to one or more environmental objectives, does no significant harm to any of the other objectives, and is carried out in compliance with minimum safeguards

* EU Regulation 2020/852 and the Delegated Acts

Key performance indicators

	2023		
(In USD million)	**Turnover**	**Capex**	**Opex**
Aligned Eligible activity	2	102	0
Eligible & not Aligned activity	12	1,085	1
Non Eligible activity	106,118	12,030	1,723
Total	**106,132**	**13,217**	**1,724**
Aligned Eligible activity	0.0%	0.8%	0.0%
Eligible & not Aligned activity	0.0%	8.2%	0.1%
Non Eligible activity	100.0%	91.0%	99.9%

	2022		
	Turnover	**Capex**	**Opex**
Aligned Eligible activity	0.0%	1.6%	0.0%
Eligible & not Aligned activity	0.0%	0.9%	0.0%
Non Eligible activity	100.0%	97.5%	100.0%

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Omnibus Simplification Package (EU)

Key Proposed Changes

Area	What's Changing
EU Taxonomy	Voluntary reporting for mid- [illegible] over [illegible] M); partial alignment accepted; simp [illegible] plate [illegible] exemptions.
CSRD	Raise reporting th [illegible] mployees; delay reporting start for Wa [illegible] voluntary reporting standard for those out [illegible]
CSDDD	Delay in i [illegible] supply chain reporting; simplify stake [illegible] arrangements.
CBAM	Exempt sma [illegible] his thresholds); optional consultation; delegate declaratio [illegible] sio [illegible] scope; adjust certificate rules.

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General example: Scatec ESG Performance Report

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PwC Assurance Statement

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https://scatec.com/wp-content/uploads/sites/7/2024/04/Scatec_ESG-Performance-Report-2023.pdf

Let's get back here

Sustainable finance typology	Value created	Factors	Optimization	Horizon
Finance-as-usual	Shareholder value	F	Max F	Short term
Sustainable Finance 1.0	Refined shareholder value	F>S and E	Max F s.t. S and E	Short term
Sustainable Finance 2.0	Stakeholder value	T=F+S+E	Max T	Medium term
Sustainable Finance 3.0	Common good value	S and E > F	Max S and E s.t. F	Long term

Source: Bruegel. Note: F = financial value; S = social impact; E =environmental impact; T = total value

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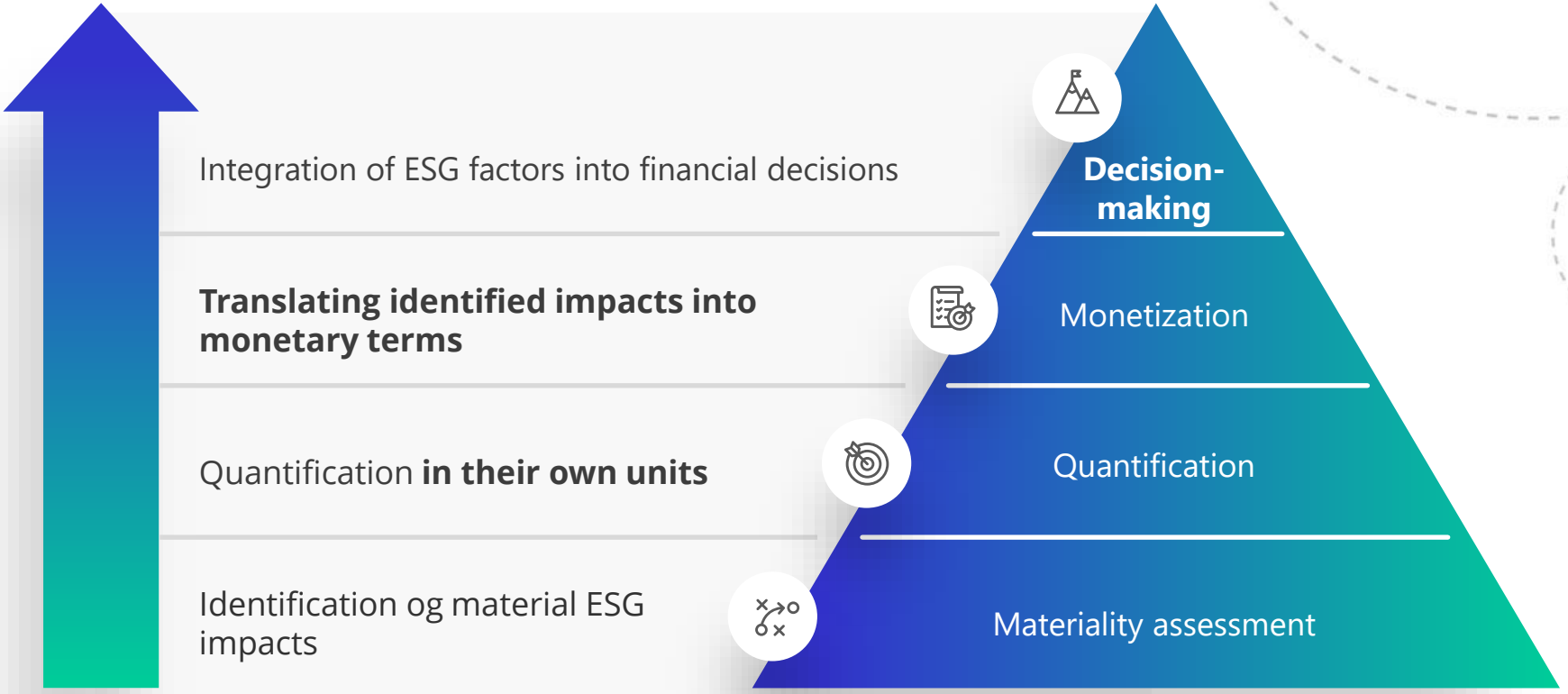
Bottom up thinking



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Can we calculate social and environmental value?

- The calculation of social value (SV) and environmental value (EV) is a recent phenomenon. This new field of impact measurement and valuation has developed methods to value social and environmental impact
- The value calculation for SV and EV can be done in three steps:
  - Materiality assessment—determine important SV and EV factors
  - Quantification—express these factors in their own units
  - Monetization—express these factors in money with shadow prices SP
- Social (S) and environmental (E) issues can be expressed in their own units then multiplied by their respective shadow price
  - life years saved by medical treatment
  - carbon emissions by using fossil fuels
  - The shadow price for one life year, for example, can be estimated and the shadow price per 1 ton of CO_2 equivalent

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Quantification

- In the second step, the S and E issues are quantified.
- For some issues that is relatively straightforward to do, especially those related to E
  - The carbon emissions of many listed airlines can now be found in their annual report.
  - It's harder to quantify for aerosol loadings and certainly harder for biodiversity
- On the social side, the value of passenger well-being is a standard indicator.
  - In such cases, estimates can be made, based, for example, on academic models or anecdotal evidence related to family, holiday, or work-related visits can be estimated, and the value of work can be expressed in living wages.
- It is possible to express components of S in their own units, such as quality life years added by a medical technology company.
  - The number of quality life years added is calculated as the change in utility value induced by the medical treatment, multiplied by the duration of the treatment effect.
- As with E, many S components are quite challenging to compute, since they lack a clear unit.
  - This applies, for example, to violations of human rights. However, organizations like Impact Institute have shown that such types of S can still be quantified, albeit less easily. This can, for example, be done by expressing the components in terms of health effects or underpayment.

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Monetizartion: Shadow prices

- Components of E and S are often hard to quantify in their own units, but can nevertheless be monetized.
  - For example, human rights violations can be expressed in monetary damages (by assessing how they hurt people's ability to lead a decent life) without taking the intermediate step of expressing them in comparable units.
  - This can, for example, be done by expressing the components in terms of health effects or underpayment.
- Welfare-Based Shadow Prices
  - The shadow prices (also called monetisation factors by the Impact Economy Foundation) should reflect the 'true scarcity' of resources to stay within planetary boundaries; or the 'true price' of human rights breaches to stay within social boundaries.
  - The term shadow prices illustrates that these prices don't reflect current market prices but 'shadow' true prices.
  - The Impact Economy Foundation (2022) and True Price (2021) provide a regularly updated list of impacts and shadow prices for a whole range of social and environmental impacts.
  - https://impacteconomyfoundation.org/impactweightedaccountsframework/ and https://trueprice.org/monetisation-factors-for-true-pricing/

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Some attempts at pricing: Case of Ambuja Cements

Figure 18 The social and environmental costs and benefits of ACL

https://trueprice.org/monetisation-factors-for-true-pricing/

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How to value E and S?

- A commonly used model is the discounted cash flow (DCF) model, which derives the value V of a project or a company as follows:

$$V = \sum_{n=0}^{N} \frac{CF_n}{(1+\rho)^n}$$

- we can calculate social value SV and environmental value EV using the standard DCF model?

$$SV = \sum_{n=0}^{N} \frac{SVF_n}{(1+\rho_s)^n} \qquad EV = \sum_{n=0}^{N} \frac{EVF_n}{(1+\rho_E)^n}$$

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Discounting Social and Environmental Capital

- Should current and future generations be treated equally? What is the appropriate discount rate for society (the social discount rate)?
  - Many moral philosophers argue for an equal treatment.
  - In economic terms, this means that the well-being of future generations gets the same weight as the well-being of the present generation => Zero-time preference
- Ramsey (1928) derives the social discount rate rs for the appraisal of social projects:

$$r_s = \delta + \eta g$$

  - δ reflects the time preference between current and future generations
  - g is growth in consumption as well as by total factor productivity growth (i.e., innovations increase efficiency of production).
  - η the elasticity of marginal utility of consumption. The elasticity measures how utility changes with consumption.

	Social discount rate $r^s = \delta + \eta \cdot g$ with $g = 1.3\%$		
Author	Time preference δ	Elasticity η	Discount rate r^s
Cline (1992)	0%	1.5	1.95%
Nordhaus (1994)	3%	1	4.3%
Stern (2006)	0.1%	1	1.4%

Source: Dasgupta (2021)

- More broadly, low social discount rates imply that future environmental and social capital is almost as scarce as current environmental and social capital. Companies that add to these scarce capitals create integrated value and will be rewarded, while companies that draw these capitals down will face an increasing cost.

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What about risk?

- The Ramsey rule measures the willingness to pay for a sure transfer of consumption through time. The risk premium for the social discount rate ?
  - Why is it problematic to use the Capital Asset Pricing Model to determine discount rates for public projects?
- There are several sources of risk related to the growth factor:
  - Growth risk: the macroeconomic risk that the growth rate of consumption fluctuates;
  - Company risk: the correlation between company risk and growth risk;
  - Catastrophe risk: the extreme element of macroeconomic risk of rare disasters (deep recessions) or society collapse.
- Risk Premium?
  - A risk premium can lead to undervaluing long-term projects that benefit future generations by focusing too much on short-term gains.

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From cashflows to investment decisions

- Standard NPV
  - NPV on F gives financially valuable projects
- Constrained PV
  - Add: S and/or E in their own units as a budget constraint
    - E: Net zero CO2 emissions
    - S: Positive health effects
- Expanded PV
  - Add: SV and/or EV in monetary terms
  - EV: CO2 emissions x price
  - SV: Positive health effects x price
- Integrated PV
  - Add: FV + SV + EV all in monetary terms
  - Integrated PV = FV + b * SV + c * EV, with b, c > 0

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Example 1: Comparing projects on constrained PV

- Suppose a medical technology company has the goal of being carbon neutral and wants all of its investment projects to contribute to that goal. The company can choose from three projects:

Comparing projects on constrained PVs

Project	Investment, € millions	NPV F, € millions	CO_2 emitted, millions	CO_2 stored, millions	NPV ≥ 0?	Contribution to CO_2 emissions ≤0?
A	70	–50	0	1	No	Yes
B	100	200	0.2	0	Yes	No
C	20	250	0.2	0	Yes	No

Comparing combinations of projects on constrained PVs

Project	Investment, € millions	NPV F, € millions	CO_2 emitted, millions	CO_2 stored, millions	NPV≥0?	Contribution to CO_2 emissions ≤0?
A + B	170	150	0.2	1	Yes	Yes
A + C	90	200	0.2	1	Yes	Yes

Example: IPV

- Let's take a steel company that is considering a project to produce a new type of steel. This project employs a carbon-intensive technology and has a productive lifetime of 5 years. The financial discount rate is 6% per year and the social discount rate is 2% per year. Table below sets out the cash flow profile, and the carbon emissions, from the project:

Year	2023	2024	2025	2026	2027	2028
Cash flows, in $ millions	-100	40	40	40	40	40
Carbon emissions, in thousands tons	0	30	30	30	30	30

- The project requires an initial investment of $100 million and then yields $40 million for the next 5 years. The carbon emissions of the production installation amount to 30,000 tons of CO2 equivalent per year.

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Solution IPV

- The first step is to calculate the environmental value flows. Assume shadow carbon price of $224 per 1 ton of CO2 equivalent (that we assume to be constant over time)

Year	2023	2024	2025	2026	2027	2028
Cash flows, in $ millions	-100	40	40	40	40	40
Carbon emissions, in thousand	0	30	30	30	30	30
Carbon price, in $	224	224	224	224	224	224
Environmental value flows, in $ millions	0	-6,72	-6,72	-6,72	-6,72	-6,72

- $V = \sum_{n=1}^{5} \frac{40}{(1+0.06)^n} - 100 = 68.5 \quad EV = \sum_{n=1}^{5} \frac{-6,72}{(1+0.02)^n} = -31.7$
- Total integrated value is 36.8 (assuming the weight to EV =1)

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Lessons from IPV model

- The IPV model acknowledges the interrelationships between the different types of value and allows a structured balancing of stakeholder interests.
  - Possibility to adjust the weights
- Flexible enough to adjust for internalizations, e.g. carbon taxes
  - Choice of weights depends not only on a company's purpose and mission, but also on the speed of internalization of negative impacts.
- The IPV model allows companies to choose their degree of sustainability.
- But there are limits
  - Dependency on shadow prices, the choice is subject to interpretation
  - Availability of company data on social and environmental impacts.

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Extra material: Bonds in all colors and shapes

- Recent increase in investors' ESG awareness
- While most investors are now familiar with green bonds, they are by no means the only way to gain exposure to ESG-labelled bonds.
- Green social and sustainability (GSS) bonds or Positive impact bonds (PIB) or ESG bonds … same thing

  ….take advantage of the demand for sustainable investments

  - Most ESG bonds are characterized by their targeting ESG-friendly projects with their proceeds: the issuer itself does not necessarily need to have strong ESG credentials.
  - A market is self-labelled: following governance and reporting practices set by bodies such as the International Capital Markets Association (ICMA) is voluntary.

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Extra material: Bonds in all colors and shapes

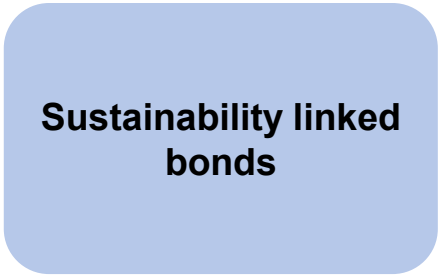
GSS+ bonds (or ESG bonds)

Use of proceeds

Issuer sustainability targets

Environmental targets

Social targets



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Summary

- Sustainable development that meets the needs of the present without compromising the ability of future generations to meet their own needs.
- Sustainable development envisions the transition from shareholder value to stakeholder value (or triple bottom line: people, planet, profit), where the final stage looks at the creation of common good value.
- To use current financial model for sustainable development externalities associated with sustainabilities need to be internalized
- Some policy instruments are already in place (e.g. carbon taxes), but it is not enough
- Companies need to explicitly integrate sustainability flows in their decisions
- There are inherent links between finance and sustainability
  - Financial industry allocates capital
  - Reaching planetary boundaries and reducing social value affect risk and return (e.g., climate and transition risks)
- Efficient allocation of capital to green project is challenging due to issues with reporting and quantification of impact
  - Several regulatory initiatives attempt to bridge the gap (EU Taxonomy)
- Moving from evaluating financial value to integrated value requires efforts and accountability, as well as proper estimation techniques with the right data sources
- In IPV approaches, F, S, and E all weigh in and can be prioritized—ideally informed by the company's purpose and value creation profile.

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